

Homework 3

601.482/682 Deep Learning

Fall 2025

September 17, 2024

Due 11:59pm EST on September 24, 2025
Please type answers inline using this LaTeX file
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Important: Please use the natural logarithm (base e) for all logarithm calculations ($\log$) below. Please show each step of your calculations; points will be deducted if only final results are present.

1. We have presented a matrix back propagation example in class. In this exercise, you will follow the same logic to derive $\frac{\partial L}{\partial X} = W^T \frac{\partial L}{\partial Y}$.
 - (a) Please derive $\frac{\partial L}{\partial W} = \frac{\partial L}{\partial Y} X^T$ (please consider the **general case** and show each step of your computation).

Let $Y = WX$, then $Y_{ij} = \sum_k W_{ik} X_{kj}$

Relationship between W , X , Y , and L : $W \xrightarrow{Y=WX} Y \xrightarrow{loss} L$

We want to find $\frac{\partial L}{\partial W_{ab}} = \sum_i \sum_j \frac{\partial L}{\partial Y_{ij}} \frac{\partial Y_{ij}}{\partial W_{ab}}$

Since $Y_{ij} = \sum_k W_{ik} X_{kj}$, $\frac{\partial Y_{ij}}{\partial W_{ab}} = \begin{cases} X_{kj}, & \text{if } i = a \text{ and } k = b, \\ 0, & \text{otherwise} \end{cases}$

$= \begin{cases} X_{bj}, & \text{if } i = a, \\ 0, & \text{otherwise} \end{cases}$

Go back to: $\frac{\partial L}{\partial W_{ab}} = \sum_i \sum_j \frac{\partial L}{\partial Y_{ij}} \cdot \underbrace{\frac{\partial Y_{ij}}{\partial W_{ab}}}_{\text{zero unless } i=a}$

$= \sum_j \frac{\partial L}{\partial Y_{aj}} X_{bj}$

We can look at the (a, b)th entry of $\left(\frac{\partial L}{\partial Y}\right) X^T$:

$$\begin{aligned} \left[\left(\frac{\partial L}{\partial Y}\right) X^T\right]_{ab} &= \sum_j \left[\frac{\partial L}{\partial Y}\right]_{aj} [X^T]_{jb} \\ &= \sum_j \frac{\partial L}{\partial Y_{aj}} X_{bj} \text{ since } X_{jb}^T = X_{bj} \end{aligned}$$

So $\frac{\partial L}{\partial W_{ab}} = \left[\left(\frac{\partial L}{\partial Y}\right) X^T\right]_{ab}$

$\therefore \frac{\partial L}{\partial W} = \left(\frac{\partial L}{\partial Y}\right) X^T$

- (b) Suppose we leverage the L_2 loss (also known as *Squared Error Loss*) for a linear regression model $Y = WX$. Given the following initialization of W and X , please calculate the updated W after one iteration. (step size = 0.1)

$$W = \begin{pmatrix} 0.3 & 0.5 \\ -0.2 & 0.4 \end{pmatrix}, X = (\mathbf{x}_1, \mathbf{x}_2) = \begin{pmatrix} 0 & 2 \\ 3 & 1 \end{pmatrix}, \hat{Y} = (\hat{\mathbf{y}}_1, \hat{\mathbf{y}}_2) = \begin{pmatrix} 0.5 & 1 \\ 1 & -1.5 \end{pmatrix}.$$

Hint: L_2 loss is defined by $L_2(Y, \hat{Y}) = (y_{11} - \hat{y}_{11})^2 + (y_{12} - \hat{y}_{12})^2 + (y_{21} - \hat{y}_{21})^2 + (y_{22} - \hat{y}_{22})^2$, where $\hat{Y}$ is the ground truth, and the model output $Y = WX$.

$$\begin{aligned} Y = WX &= \begin{pmatrix} 0.3 \cdot 0 + 0.5 \cdot 3 & 0.3 \cdot 2 + 0.5 \cdot 1 \\ -0.2 \cdot 0 + 0.4 \cdot 3 & -0.2 \cdot 2 + 0.4 \cdot 1 \end{pmatrix} \\ &= \begin{pmatrix} 1.5 & 1.1 \\ 1.2 & 0 \end{pmatrix} \end{aligned}$$

Since L_2 loss is defined by $L_2(Y, \hat{Y}) = \sum_i \sum_j (y_{ij} - \hat{y}_{ij})^2$:

$$\begin{aligned} \frac{\partial L}{\partial Y} &= 2(Y - \hat{Y}) \\ &= 2 \begin{pmatrix} 1.5 - 0.5 & 1.1 - 1 \\ 1.2 - 1 & 0 - (-1.5) \end{pmatrix} \\ &= 2 \begin{pmatrix} 1 & 0.1 \\ 0.2 & 1.5 \end{pmatrix} \\ &= \begin{pmatrix} 2 & 0.2 \\ 0.4 & 3 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \frac{\partial L}{\partial W} &= \left(\frac{\partial L}{\partial Y} \right) X^T \text{ as we found in part a)} \\ &= \begin{pmatrix} 2 & 0.2 \\ 0.4 & 3 \end{pmatrix} \begin{pmatrix} 0 & 3 \\ 2 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 0.4 & 6.2 \\ 6 & 4.2 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} W_{new} &= \begin{pmatrix} 0.3 & 0.5 \\ -0.2 & 0.4 \end{pmatrix} - 0.1 \begin{pmatrix} 0.4 & 6.2 \\ 6 & 4.2 \end{pmatrix} \\ &= \begin{pmatrix} 0.26 & -0.12 \\ -0.8 & -0.02 \end{pmatrix} \end{aligned}$$

2. In this exercise, we will explore how vanishing and exploding gradients affect the learning process. Consider a simple 3-layer network with 1-dimensional data $x \in \mathbb{R}$, prediction $y \in [0, 1]$, true label $\hat{y} \in \{0, 1\}$, and weights $w_1, w_2, w_3 \in \mathbb{R}$. Weights are initialized randomly from Standard Gaussian i.e., $w_1, w_2, w_3 \sim \mathcal{N}(0, 1)$. We use the sigmoid activation function σ between all layers. First, consider a classification task with the cross entropy loss function $L(y, \hat{y}) = -(\hat{y} \log(y) + (1 - \hat{y}) \log(1 - y))$. Formally, the network can be represented as: $y = \sigma(w_3 \cdot \sigma(w_2 \cdot \sigma(w_1 \cdot x)))$. Note that for this problem, we exclude bias terms for simplicity; but in a general case, you should always consider the bias terms. *Hint:* Sigmoid is defined by $\sigma(x) = \frac{1}{1+e^{-x}}$.

- (a) Compute the derivative for a sigmoid function σ . What are the maximum extrema and minimum extrema of this derivative, and when are they reached?

$$\begin{aligned} \sigma'(x) &= -(1 + e^{-x})^{-2}(-e^{-x}) \\ &= \frac{e^{-x}}{(1+e^{-x})^2} \end{aligned}$$

$$\begin{aligned}
&= \left(\frac{1}{1+e^{-x}}\right)\left(\frac{e^{-x}}{1+e^{-x}}\right) \\
&= \left(\frac{1}{1+e^{-x}}\right)\left(\frac{1+e^{-x}-1}{1+e^{-x}}\right) \\
&= \left(\frac{1}{1+e^{-x}}\right)\left(\frac{1+e^{-x}}{1+e^{-x}} - \frac{1}{1+e^{-x}}\right) \\
&= \sigma(x) \cdot \left(1 - \frac{1}{1+e^{-x}}\right) \\
&= \sigma(x) \cdot (1 - \sigma(x))
\end{aligned}$$

Let $f[\sigma(x)] = \sigma(x) \cdot (1 - \sigma(x))$

Set derivative of f to zero to find the critical point: $\frac{df[\sigma(x)]}{d\sigma(x)} = 1 - 2\sigma(x) = 0$

$$\sigma(x) = \frac{1}{2}$$

Take second derivative to check concavity: $\frac{d^2f[\sigma(x)]}{d\sigma(x)^2} = -2$ so it's concave.

Thus maximum is at $\sigma'(x) = \frac{1}{2}(1 - \frac{1}{2}) = \frac{1}{4}$. This occurs when $x = 0$.

Since $0 < \sigma(x) < 1$, both $\sigma(x)$ and $1 - \sigma(x)$ are positive, so $\sigma'(x)$ must always be positive. Thus the minimum is $\sigma'(x) \rightarrow 0$. This occurs when $x \rightarrow +\infty$ or $-\infty$.

- (b) Consider a random initialization: $w_1 = 0.25, w_2 = -0.11, w_3 = 0.78$, and one specific data sample ($x = 0.63, \hat{y} = 1$). Apply back propagation to compute the gradients for each weight. What do you notice about the magnitude?

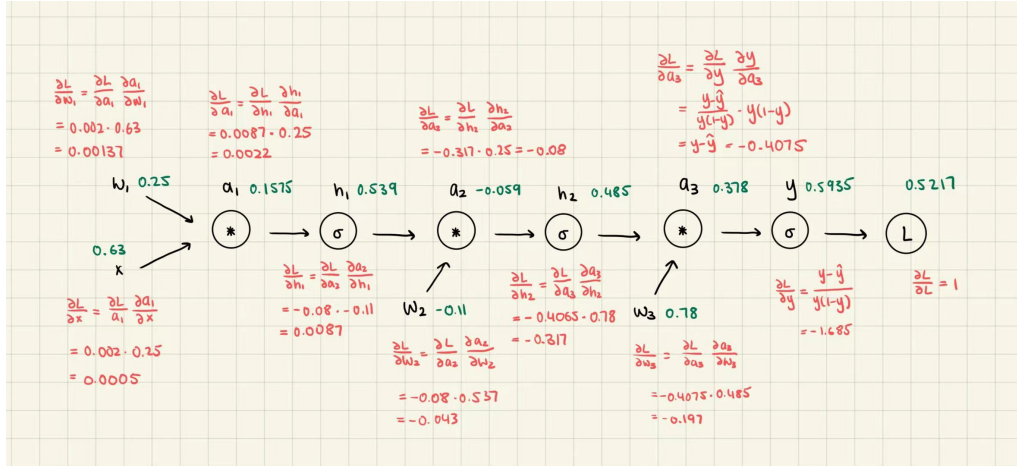


Figure 1: Computational graph.

$$\begin{aligned}
a_1 &= w_1 x, h_1 = \sigma(a_1), \\
a_2 &= w_2 h_1, h_2 = \sigma(a_2), \\
a_3 &= w_3 h_2, y = \sigma(a_3)
\end{aligned}$$

$$\begin{aligned}
a_1 &= 0.1575, h_1 = 0.53929 \\
a_2 &= -0.05932, h_2 = 0.48517, \\
a_3 &= 0.37844, y = 0.59350
\end{aligned}$$

Layer 3:

$$\begin{aligned}\frac{\partial L}{\partial y} &= -\frac{\hat{y}}{y} + \frac{1-\hat{y}}{1-y} \\ &= \frac{-\hat{y} + \hat{y}y + y - \hat{y}y}{y(1-y)} \\ &= \frac{y - \hat{y}}{y(1-y)}\end{aligned}$$

$$\begin{aligned}\frac{\partial y}{\partial a_3} &= \sigma(a_3)[1 - \sigma(a_3)] \\ &= y(1-y)\end{aligned}$$

$$\frac{\partial L}{\partial a_3} = \frac{\partial L}{\partial y} \frac{\partial y}{\partial a_3} = \frac{y - \hat{y}}{y(1-y)} \cdot y(1-y) = y - \hat{y} = 0.59350 - 1 = -0.40650$$

$$\frac{\partial L}{\partial w_3} = \frac{\partial L}{\partial a_3} \frac{\partial a_3}{\partial w_3} = \frac{\partial L}{\partial a_3} h_2 = -0.40650 \cdot 0.48517 = -0.19723$$

Layer 2:

$$\frac{\partial L}{\partial h_2} = \frac{\partial L}{\partial a_3} \frac{\partial a_3}{\partial h_2} = \frac{\partial L}{\partial a_3} w_3 = -0.40650 \cdot 0.78 = -0.31707$$

$$\frac{\partial h_2}{\partial a_2} = \sigma'(a_2) = \sigma(a_2)[1 - \sigma(a_2)] = h_2(1 - h_2) = 0.24978$$

$$\frac{\partial L}{\partial a_2} = \frac{\partial L}{\partial h_2} \frac{\partial h_2}{\partial a_2} = -0.31707 \cdot 0.24978 = -0.07919$$

$$\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial a_2} \frac{\partial a_2}{\partial w_2} = \frac{\partial L}{\partial a_2} h_1 = -0.07919 \cdot 0.53929 = -0.0427$$

Layer 1:

$$\frac{\partial L}{\partial h_1} = \frac{\partial L}{\partial a_2} \frac{\partial a_2}{\partial h_1} = \frac{\partial L}{\partial a_2} w_2 = -0.07919 \cdot -0.11 = 0.00871$$

$$\frac{\partial h_1}{\partial a_1} = \sigma'(a_1) = \sigma(a_1)[1 - \sigma(a_1)] = h_1(1 - h_1) = 0.2485$$

$$\frac{\partial L}{\partial a_1} = \frac{\partial L}{\partial h_1} \frac{\partial h_1}{\partial a_1} = 0.00871 \cdot 0.2485 = 0.00216$$

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial a_1} \frac{\partial a_1}{\partial w_1} = \frac{\partial L}{\partial a_1} x = 0.00216 \cdot 0.63 = 0.001364$$

$$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial a_1} \frac{\partial a_1}{\partial x} = \frac{\partial L}{\partial a_1} w_1 = 0.00216 \cdot 0.25 = 0.0005$$

Since we are using sigmoids (i.e. $\sigma'(a_l) \leq 0.25$), the gradients decrease rapidly as we move backwards. This is the problem of vanishing gradients:

$$\left| \frac{\partial L}{\partial w_3} \right| = 0.12 > \left| \frac{\partial L}{\partial w_2} \right| = 0.04 > \left| \frac{\partial L}{\partial w_1} \right| = 0.0014$$

Now, we switch to a regression task while keeping the same network structure. We remove the final sigmoid activation, so our new network turns to be $y = w_3 \cdot \sigma(w_2 \cdot \sigma(w_1 \cdot x))$, where predictions $y \in \mathbb{R}$ and targets $\hat{y} \in \mathbb{R}$. We use the L_2 loss function to substitute the cross entropy, i.e. $L(y, \hat{y}) = (\hat{y} - y)^2$.

(c) Derive the gradient of the loss function with respect to the weights (w_1, w_2, w_3) .

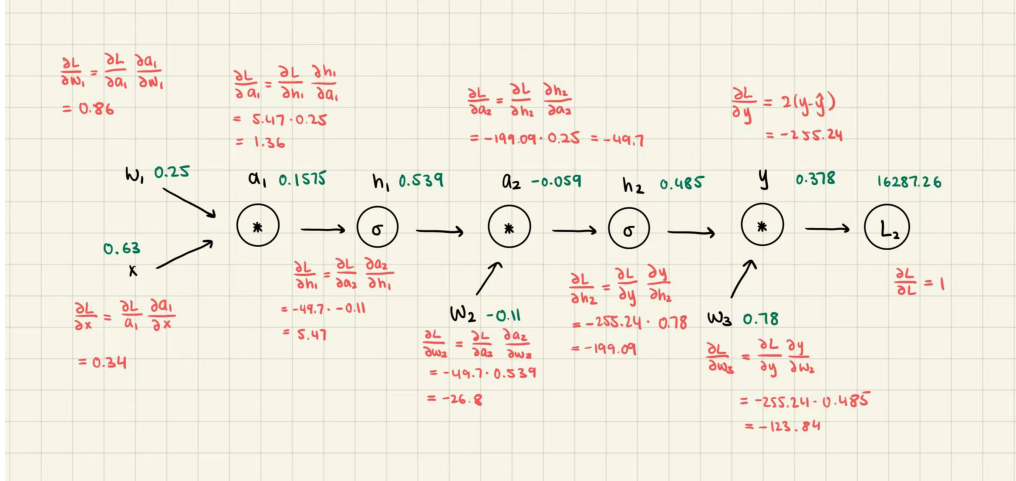


Figure 2: Computational graph.

Let $a_1 = w_1 x$, $h_1 = \sigma(a_1)$,
 $a_2 = w_2 h_1$, $h_2 = \sigma(a_2)$,
 $y = w_3 h_2$

Layer 3:

$$\frac{\partial L}{\partial y} = -2(\hat{y} - y) = 2(y - \hat{y})$$

$$\frac{\partial L}{\partial w_3} = \frac{\partial L}{\partial y} \frac{\partial y}{\partial w_3} = 2(y - \hat{y})h_2$$

Layer 2:

$$\frac{\partial L}{\partial h_2} = \frac{\partial L}{\partial y} \frac{\partial y}{\partial h_2} = 2(y - \hat{y})w_3$$

$$\frac{\partial L}{\partial a_2} = \frac{\partial L}{\partial h_2} \frac{\partial h_2}{\partial a_2} = 2(y - \hat{y})w_3 \sigma'(a_2)$$

$$= 2(y - \hat{y})w_3 [h_2(1 - h_2)]$$

$$\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial a_2} \frac{\partial a_2}{\partial w_2} = 2(y - \hat{y})w_3 [h_2(1 - h_2)]h_1$$

Layer 1:

$$\frac{\partial L}{\partial h_1} = \frac{\partial L}{\partial a_2} \frac{\partial a_2}{\partial h_1} = 2(y - \hat{y})w_3 [h_2(1 - h_2)]w_2$$

$$\frac{\partial L}{\partial a_1} = \frac{\partial L}{\partial h_1} \frac{\partial h_1}{\partial a_1} = 2(y - \hat{y})w_3 [h_2(1 - h_2)]w_2 \sigma'(a_1)$$

$$= 2(y - \hat{y})w_3 [h_2(1 - h_2)]w_2 h_1(1 - h_1)$$

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial a_1} \frac{\partial a_1}{\partial w_1} = 2(y - \hat{y})w_3 [h_2(1 - h_2)]w_2 h_1(1 - h_1)x$$

- (d) Consider again the initialization of $w_1 = 0.25$, $w_2 = -0.11$, $w_3 = 0.78$, and data sample ($x = 0.63$, $\hat{y} = 128$). Apply back propagation to compute the gradients for each weight. What do you notice about the magnitude?

$$a_1 = 0.25 \cdot 0.63 = 0.1575,$$

$$h_1 = \sigma(a_1) = 0.54,$$

$$\begin{aligned}
a_2 &= -0.11 \cdot 0.5392938 = -0.059, \\
h_2 &= \sigma(a_2) = 0.485 \\
y &= 0.78 \cdot 0.4851738 = 0.378
\end{aligned}$$

$$\frac{\partial L}{\partial w_3} = 2(y - \hat{y})h_2 = -255.244 \cdot 0.485 = -123.8$$

$$\frac{\partial L}{\partial w_2} = 2(y - \hat{y})w_3[h_2(1 - h_2)]h_1 = -255.244 \cdot 0.78 \cdot 0.25 \cdot 0.539 = -49.7 \cdot 0.539 = -26.8$$

$$\frac{\partial L}{\partial w_1} = 2(y - \hat{y})w_3[h_2(1 - h_2)]w_2h_1(1 - h_1)x = -49.7 \cdot -0.11 \cdot 0.25 \cdot 0.63 = 0.86$$

The $L_2 = (\hat{y} - y)^2$ here is huge because the target is huge (128) while the model output is small (0.278).

But since we are still using sigmoids (i.e. $\sigma'(a_l) \leq 0.25$), the gradients still decrease rapidly as we move backwards:

$$\left| \frac{\partial L}{\partial w_3} \right| = 123.8 > \left| \frac{\partial L}{\partial w_2} \right| = 26.8 > \left| \frac{\partial L}{\partial w_1} \right| = 0.86$$

Compared to part a/b, however, the gradients are much larger because the difference between $\hat{y}$ and y are huge, and theres no final sigmoid to make the final gradient even smaller.